

Project Requirement Document : Airfare & Passenger Behavior Analysis

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Data: Kaggle

Purpose:

This analysis aims to support data-driven decisions in airline management by uncovering how fare changes affect passenger volume. The project segments routes based on pricing and demand patterns, helping stakeholders optimize pricing strategies, identify profitable markets, and flag underperforming routes.

Key dependencies:

This project will require a dataset of flight fares and number of passenger travelled. The dataset obtained from an external data source contains the data that is helpful for to conduct this project.

Stakeholder requirements: (R - required, D - desired, or N - nice to have.)

In order to continuously improve and effectively market trips, the dashboard must help Airlines decision-makers understand how their customers are using the airlines and the demand at different locations, including factors that might influence that demand at different times.

- Identify trends in airfare and passenger behavior over cities R
- Evaluate the relationship between average fare and passengers volume R
- Segment markets based on fare/passenger combinations to spot growth or risk areas N
- Build business-ready visuals and insights that support pricing and route decisions D
- Build a visualization that shows fare changes over the years N
- Identify opportunities for pricing optimization and route management decisions R

Success criteria:

- Data Quality: Successfully clean, process, and prepare the dataset with no missing or inconsistent values impacting analysis.
- Actionable Segmentation: Accurately segment city routes based on fare and passenger volume into meaningful categories (e.g., premium, volume-driven, underperforming).
- Business Insights: Provide at least 3 clear business insights related to pricing strategy, route performance, or market opportunities.
- Visual Communication: Present findings through easy-to-understand charts, graphs, and dashboards that support stakeholder decision-making.

Assumptions:

- **Data Accuracy**
The airfare and passenger data provided is assumed to be accurate, complete, and reflective of actual airline operations during the recorded periods.
- **Market Stability**
No significant external shocks (e.g., pandemics, natural disasters, major policy changes) occurred during the dataset timeline that would drastically alter normal passenger behavior.
- **Price Consistency**
The average fare reported per city is assumed to represent the typical fare for that market, without considering promotional discounts, loyalty programs, or special rates.
- **Passenger Type Mix**
Passenger data does not distinguish between business and leisure travelers. The analysis assumes that fare trends affect all passenger types similarly, though real-world behavior may vary.
- **Geographical Uniformity**
All city routes are treated independently without considering connecting flights, hub-spoke models, or competition from nearby airports.
- **Yield Interpretation**
Yield (revenue per mile) is not directly analyzed, but fare is used as a proxy for pricing strategy in relation to passenger volume..

Compliance and privacy:

The data must not include any personal data such as name, email address, phone number, or physical address. The user provides this data as part of their device activation but is not necessary for this project. It is paramount that the users be anonymized to avoid any bias.

Accessibility:

The dashboards should offer text alternatives including large print and text-to-speech.

Roll-out plan:

- Data Cleaning & Preparation
- Exploratory Data Analysis
- Correlation Analysis

- Market Segmentation